METRIC

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SUPERSEDING
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DETAIL SPECIFICATION

POTASSIUM NITRATE
(For use in ammunition)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for commercial grade potassium nitrate and is intended for use in ammunition.

1.2 Classes. Potassium nitrate has the following classes:

Class 1 - for use in black powder and chemical munitions
Class 2 - for use in pyrotechnic compositions
Class 3 - for use in propellants
Class 4 - for use in benite, CBI Type I, M2, M5, M8, M26, and M30A2 propellants

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in section 3 or 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: US Army Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000 or email to usarmy.pica.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil/.

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2.2 Government documents.

2.2.1 Specification and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN CHEMICAL SOCIETY

Reagent Chemicals	-	Specifications and Procedures for Reagents and Standard Grade Reference Materials

(Copies of this document are available from https://pubs.acs.org/.)

ASTM INTERNATIONAL

ASTM E11	-	Standard Specification for Wire Cloth and Sieves for Testing Purposes
ASTM E70	-	Standard Test Method for pH of Aqueous Solutions with the Glass Electrode
ASTM E300	-	Standard Practice for Sampling Industrial Chemicals
ASTM D1193	-	Standard Specification for Reagent Water

(Copies of these documents are available from https://www.astm.org.)

2.4 Order of precedence. Unless otherwise specified in this document or in the contract, in the event of a conflict between the text of this document and references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

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3.1.1 First article inspection. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subjected to conformance inspection in accordance with 4.3.

3.2 Physical and chemical properties. The potassium nitrate shall comply with the physical and chemical requirements specified in Table I.

TABLE I. Physical and chemical properties.

Property	Requirement			
	Class 1	Class 2	Class 3	Class 4
Moisture, weight percent maximum (wt. % max.)	0.2	0.2	0.2	0.2
pH	7.0±1.0	7.0±1.0	7.0±1.0	7.0±1.0
Grit	None	None	None	2/
Insoluble material, wt. % max.	0.1	0.1	0.1	0.1
Nitrogen, wt. % minimum (min.)	13.77	13.77	13.77	13.77
Chlorides, as potassium chloride (KCl), wt. % max.	0.07	0.07	0.07	0.07
Chlorates and perchlorates, as potassium perchlorate ($KClO_4$), wt. % max.	0.5	0.5	0.5	0.5
Iron and aluminum, as oxides, wt. % max.	0.5	0.5	-----	-----
Calcium and magnesium, as oxides, wt. % max.	0.5	0.5	0.5	0.5
Sodium, as oxide, wt. % max.	0.25	0.25	-----	-----
Ammonium compounds	None	None	-----	-----
Odor	None	None	None	None
Color	White	White	White	White
Notes: 1/ "-----" indicates no established requirement. 2/ Requirement for Class 4 is insoluble material including grit.				

3.3 Granulation. The granulation requirements shall be as follows.

3.3.1 Class 1. Class 1 potassium nitrate shall be furnished in lump form.

3.3.2 Classes 2, 3, and 4. Classes 2, 3, and 4 potassium nitrate shall comply with the granulation requirements shown in Table II.

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TABLE II. Granulation requirements (Classes 2, 3, and 4).

U.S. Alternate Sieve Number	Weight percent through specified sieve, minimum	
	Class 2	Classes 3 and 4
60	99.9	-----
80	-----	99.9
100	40	-----
Notes: 1/ U.S. Alternative Sieves shall be in accordance with ASTM E11.		

3.4 Workmanship. The potassium nitrate shall be uniform in quality and free from dust, debris, visible impurities, or foreign matter.

4. VERIFICATION

TABLE III. Requirement/verification cross reference matrix.

Method of Verification 1 – Analysis 2 – Demonstration (end item test) 3 – Examination 4 – Test			Classes of Verification A – First Article Inspection B – Conformance Inspection					
Section 3 Requirements	Description	Section 4 Verifications	Verification Method				Verification Class	
			1	2	3	4	A	B
3.1.1	First article inspection	4.2	X		X	X	X	
3.1.2	Conformance	4.3	X		X	X		X
3.2	Moisture	4.4.1	X			X	X	X
3.2	pH	4.4.2	X			X	X	X
3.2	Grit	4.4.3	X			X	X	X
3.2	Insoluble material	4.4.4	X			X	X	X
3.2	Nitrogen	4.4.5	X			X	X	X
3.2	Chlorides	4.4.6	X			X	X	X
3.2	Chlorates and perchlorates	4.4.7	X			X	X	X
3.2	Iron and aluminum	4.4.8	X			X	X	X
3.2	Calcium and magnesium	4.4.8	X			X	X	X
3.2	Sodium	4.4.8	X			X	X	X
3.2	Ammonium compounds	4.4.9	X			X	X	X
3.2	Odor	4.4.11			X		X	X
3.2	Color	4.4.11			X		X	X
3.3	Granulation	4.4.10	X			X	X	X
3.4	Workmanship	4.4.11			X		X	X

4.1 Classification of inspection. The verification requirements specified herein are classified as follows.

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a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 First article inspection. When specified (see 6.2), a sample shall be subjected to first article verification in accordance with Table IV. Sampling procedures shall be conducted in accordance with 4.3.2. Samples shall be produced by the contractor using the same process procedures, material supply sources, production equipment, and production line as will be used in regular production.

4.2.1 Rejection. The first article sample shall be rejected if any of the inspections/tests in Table IV fails to meet any of the requirements.

TABLE IV. First article inspection.

Category	Examination or test	Requirement paragraph	Inspection method reference
Critical	None defined		
Major			
101	Moisture	3.2	4.4.1
102	pH	3.2	4.4.2
103	Grit	3.2	4.4.3
104	Insoluble material	3.2	4.4.4
105	Nitrogen	3.2	4.4.5
106	Chlorides	3.2	4.4.6
107	Chlorates and perchlorates	3.2	4.4.7
108	Iron and aluminum	3.2	4.4.8.1 or 4.4.8.2
109	Calcium and magnesium	3.2	4.4.8.1 or 4.4.8.3
110	Sodium	3.2	4.4.8.1
111	Ammonium compounds	3.2	4.4.9
112	Odor	3.2	Smell/4.4.11
113	Coloration	3.2	Visual/4.4.11
114	Granulation	3.3	4.4.10
Minor			
201	Workmanship	3.4	Visual/4.4.11

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Inspection lots and batches shall comply with the MIL-STD-1916, paragraph “Formation and identification of lots or batches.” A lot shall consist of one or more acceptable batches of potassium nitrate produced by one manufacturer in accordance with one specification and revision under one continuous set of operating. Each batch shall consist of that quantity of potassium nitrate that has been subjected to the same chemical and physical unit operations intended to make the final product homogeneous.

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4.3.2 Potassium nitrate sampling. Sampling for inspection shall be performed at either the supplier location where the potassium nitrate is packaged, or at the receiving facility where the material is unpacked for use in the next higher sub-assembly or formulation.

4.3.2.1 Batch sampling at the supplier. Samples shall be collected from the beginning, middle, and end of every batch in every lot in accordance with ASTM E300. A composite sample of not less than 500 grams shall be formed which is assumed to be representative of that batch. All composite samples shall be verified in accordance with 4.4. If any sample fails to comply with any applicable requirement the batch shall be rejected.

4.3.2.2 Sampling at the receiving facility. Potassium nitrate shall be sampled using Verification Level II of MIL-STD-1916, based upon the number of shipping containers (e.g., drums, boxes, etc.) in a lot. Each shipping container shall be sampled in three locations in accordance with ASTM E300 using a thief stick. The samples from each shipping container shall be placed in a secondary airtight container and mixed to form a composite sample of not less than 500 g for each potassium nitrate lot received. There shall be no reduced sampling. Every lot received shall be tested in accordance with 4.3.3. If any sample fails to meet any test requirement, the batch/lot represented by the sample shall be rejected.

4.3.3 Potassium nitrate – classification of characteristics.

Category	Examination or test	Verification level	Requirement paragraph	Inspection method
Critical	None			
Major				
101	Moisture	4.3.2	3.2	4.4.1
102	pH	4.3.2	3.2	4.4.2
103	Grit	4.3.2	3.2	4.4.3
104	Insoluble material	4.3.2	3.2	4.4.4
105	Nitrogen	4.3.2	3.2	4.4.5
106	Chlorides	4.3.2	3.2	4.4.6
107	Chlorates and perchlorates	4.3.2	3.2	4.4.7
108	Iron and aluminum	4.3.2	3.2	4.4.8.1 or 4.4.8.2
109	Calcium and magnesium	4.3.2	3.2	4.4.8.1 or 4.4.8.3
110	Sodium	4.3.2	3.2	4.4.8.1
111	Ammonium compounds	4.3.2	3.2	4.4.9
112	Odor	4.3.2	3.2	Smell/4.4.11
113	Coloration	4.3.2	3.2	Visual/4.4.11
114	Granulation	4.3.2	3.3	4.4.10
Minor				
201	Workmanship	4.3.2	3.4	Visual/4.4.11

4.4 Test methods and procedures. Unless otherwise specified, all chemicals and reagents shall be reagent grade or better, and water shall be distilled, deionized, or equivalent, conforming to ASTM D1193 Type I or Type II.

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4.4.1 Moisture. Moisture content shall be determined in accordance with the following method. Grind a portion of sample to the consistency of table salt (easily passes through a U.S. Alternative No. 40 sieve). Accurately weigh approximately 10.000 g of the sample. Dry in an oven at 150 ± 2 degrees Celsius (°C) for not less than two hours and reweigh. Continue drying with hourly checks until constant weight is reached as shown by readings of not greater than two (2) mg difference. Cool in a desiccator and record final weight. Calculate the loss in weight as percent moisture.

$$\text{Percent moisture} = 100 \times \left(\frac{A - B}{A}\right)$$

Where: A = original sample weight, g
B = final dried sample weight, g

4.4.2 pH. pH shall be determined in accordance with either of the following methods.

4.4.2.1 pH by meter. Weigh 5 ± 0.05 g of the sample and place in a 250 milliliter (mL) beaker. Add 100 mL of neutral distilled water to the beaker and dissolve the sample. Determine the pH of this solution in accordance with ASTM E70.

4.4.2.2 pH by litmus paper (alternate). Dissolve a weighed portion of 5 ± 0.05 g of the sample in 20 mL of neutral distilled water and determine if sensitive blue litmus paper is changed in color within 3 minutes, when immersed in the solution. The blue litmus paper shall be considered to be sufficiently sensitive if it shows a distinct reddish coloration, after being immersed for 3 minutes in 0.0001 normal (N) hydrochloric or sulfuric acid.

Retaining the solution from the previous acidity test determine if sensitive red litmus paper is changed in color within 3 minutes, when immersed in the solution. The red litmus paper shall be considered sensitive enough if it shows a distinct bluish coloration, after being immersed for 3 minutes in 0.0001 N sodium hydroxide.

The sample shall be considered neutral if neither litmus paper changes color.

4.4.3 Grit. Grit shall be determined in accordance with the following method. Dissolve an approximately 25 g portion of the sample in water, filter through a medium porosity filter, and wash thoroughly. Transfer any material retained on the filter to a smooth glass slide. Press and rub the material with a smooth steel spatula against the glass slide. The presence of grit is indicated by the presence of a scratching noise.

4.4.4 Insoluble material (and grit, Class 4 only). Insoluble material shall be determined in accordance with the following method. Dissolve a weighed portion of approximately 10 g of the sample in 100 mL of boiling distilled water and filter the solution through a tared filtering crucible of medium porosity. Wash the insoluble residue on the filter with hot water. Dry the crucible and contents at 100°C to 105°C for 3 hours, cool in a desiccator, and weigh. Calculate the increase in weight as percent insoluble material.

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$$\text{Percent insoluble material} = 100 \times \left(\frac{A - B}{C}\right)$$

Where: A = weight of residue and crucible, in g
B = tared weight of crucible, in g
C = weight of sample, in g

4.4.5 Nitrogen. Nitrogen content shall be determined in accordance with either of the following methods.

4.4.5.1 Determination of nitrogen by ion chromatography. The following procedure is used to determine nitrogen content of potassium nitrate. The potassium nitrate sample is dissolved in deionized water and the solution analyzed for nitrate by suppressed anion chromatography with conductivity detection. The nitrate content is determined by comparison to a National Institute of Standards and Technology (NIST) reference standard.

4.4.5.1.1 Reagents.

a. Deionized or distilled water.
b. Reference potassium nitrate (NIST standard reference material (SRM) 193 has been used and found acceptable).
c. Chromatography eluent, an aqueous solution in deionized water, typically containing 2 to 6 millimolar sodium carbonate and 1 to 2 millimolar sodium bicarbonate. The exact ratio depends on chromatography column (see 4.4.5.1.2.d); a low concentration of carbonate yields slower elution but better resolution while a high concentration of carbonate yields faster elution but poorer resolution.
d. Suppressor regenerating solution as recommended by chromatograph manufacturer, typically a 50-100 millimolar solution of sulfuric acid in deionized water.

4.4.5.1.2 Supplies and equipment.

a. Analytical balance, 200 gram capacity, 0.0001 gram resolution or better.
b. Oven, set to 150˚C ± 2˚C.
c. Ion chromatograph with a 20 microliter fixed-volume loop injector, anion suppression, and conductivity detector with minimum sensitivity of 100 microsiemens/centimeter full scale.
d. Ion chromatography column, which together with the proper eluent (4.4.5.1.1.c) can baseline-resolve nitrate and completely elute sulfate in 30 minutes or less, e.g. Metrohm Metrosep ASupp 5 150 x 4 millimeters 5 micron, or equivalent.
e. Volumetric flasks, quantity: 3 x 1 L, 1-3 x 250 mL, Class A.
f. Volumetric pipet, quantity: 1 x 5 mL, Class A.
g. Prior to starting, all flasks shall be rinsed at least three (3) times with deionized water and allowed to air dry.

4.4.5.1.3 Calibration standards. Calibration standards shall be prepared as follows:

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a. Crush 5 ± 1 g of the reference potassium nitrate (4.4.5.1.1.b) to pass through a U.S. Alternative No. 100 sieve conforming to ASTM E11, and dry for 2 hours in a 150˚C ± 2˚C oven. Cool in a desiccator.

b. Weigh 0.8150 ± 0.0050 g, and record exact weight, of the dried and cooled potassium nitrate into a 250 mL volumetric flask. Fill to about half volume with deionized water and swirl to dissolve. Make up to volume with deionized water and mix well.

c. Fill a 5 mL volumetric pipet with the reference stock solution (4.4.5.1.3.b) and discard to waste. Fill and discard once more.

d. Fill the pipet with a new 5 mL aliquot of the reference stock solution (step b) and transfer to a 1 L volumetric flask. After bulk of the solution drains into the flask, wait 10-15 seconds for the last few drops to gravity-drain, but do not blow out the remainder in the pipet tip. Make the flask up to volume with deionized water and mix well.

e. Rinse the pipet twice with 5 mL of deionized water and discard the washings. Use the pipet to dilute the samples (4.4.5.1.4.b).

4.4.5.1.4 Sample preparation. The nitrogen analysis shall be performed on two replicate aliquots, 0.815 gram each, of potassium nitrate.

a. Crush 5 ± 1 g of the sample potassium nitrate to pass through a U.S. Alternative No. 100 sieve conforming to ASTM E11, and dry for 2 hours in a 150˚C ± 2˚C oven. Cool in a desiccator.

b. Weigh 0.8150 ± 0.0050 g, and record exact weight, of the dried and cooled sample into a 250 mL volumetric flask. Fill to about half volume with deionized water and swirl to dissolve. Make up to volume with deionized water and mix well.

c. Using the same 5 mL volumetric pipet that was used to prepare the standard solution (4.4.5.1.3.e), fill with the sample solution and discard to waste. Fill and discard once more.

d. Fill the pipet with a new 5 mL aliquot of the sample solution and transfer to a 1 L volumetric flask. Use same transfer technique as for preparing the standard solution in 4.4.5.1.3.d.

e. Rinse the pipet twice with 5 mL of deionized water and repeat 4.4.5.1.4.b-d to prepare the second replicate.

f. Make both 1 L flasks up to volume with deionized water and mix well.

4.4.5.1.5 Procedure.

a. Set up the ion chromatograph in accordance with manufacturer instructions. The instrument shall be equilibrated until a steady, linear baseline is obtained. It is recommended to inject 1-3 samples of deionized water while the instrument is equilibrating to make sure the chromatograms do not contain any analytes other than water and/or system peaks.

b. Fill up appropriate instrument vials, enough to make three (3) injections of the reference solution (4.4.5.1.3), and two (2) injections each of both sample replicates (4.4.5.1.4.f).

c. All solutions shall be analyzed using a method appropriate for the selected chromatography column (4.4.5.1.2.d) and system.

d. Structure the injection sequence to make two (2) injections of the reference solution, followed by two (2) injection each of the sample solutions, followed by one (1) injection of the

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reference solution. Allow each run to be just long enough (30 minutes minimum) to completely elute any sulfate which may interfere with integration of the subsequent sample.

e. Identify the nitrate peak, which elutes around the middle of each run. For the standard, nitrate shall be the largest peak in the chromatogram; note its retention time. For the samples, examine the same retention time ± 10%; nitrate shall be identified as the largest peak in the same retention time window.

f. Integrate each chromatogram using valley-to-valley type integration making sure the start and end points of the nitrate peak both coincide with the local baseline minima. Obtain the nitrate peak areas for each of the seven runs.

g. For the three reference solution injections, obtain the arithmetic mean of the nitrate areas.

h. For each sample solution, obtain the arithmetic mean of the nitrate peak areas for the two injections thereof.

i. Calculate nitrogen content for each sample as follows:

$$\% \text{ Nitrogen} = \frac{13.85 \times A_s \times W_{ref}}{A_{ref} \times W_s}$$

Where: A_S = average nitrate area, sample (4.4.5.1.5.h)
W_{REF} = reference weight, g (4.4.5.1.3.b)
A_{REF} = average nitrate area, reference (4.4.5.1.5.g)
W_S = sample weight for replicate, g (4.4.5.1.4.b)

j. Calculate and report the arithmetic mean of the two sample replicates.

4.4.5.2 Determination of nitrogen by ferrous sulfate titration method (alternate).

4.4.5.2.1 Reagents. The following chemicals shall be used:

a. Sulfuric acid (H_2SO_4), 95-96%

b. Reference potassium nitrate (NIST standard reference material (SRM) 193 has been used and found acceptable).

c. Ferrous sulfate ($FeSO_4$), 0.4% nitrogen per milliliter. Dissolve 350 g of $FeSO_4$ 7•H_2O (ACS grade) in 800 mL distilled water. Slowly add one liter cold 1:1 H_2SO_4 and dilute to two liters with distilled water. Preserve titrant with nitrogen purge. Allow solution to stand for minimum three days.

4.4.5.2.2 Apparatus and supplies.

a. Fisher Titrimeter II System with controller, pH/redox meter, buret-dispenser with 25 mL amber syringe, stirrer, recorder, and platinum and glass electrodes, or equivalent.

b. Nitrogen purge on titrant reservoir, 2 ounce per square inch

c. Circulating-cooling bath and a small external bath with coolant maintained between 2-5˚C

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4.4.5.2.3 Standardization of the ferrous sulfate. Perform standardization in duplicate. Duplicates shall agree within 0.0005 (see factor F in equation below). If duplicates do not agree re-run or prepare new standard solutions until agreement is achieved.

Note: if the system has been down for eight hours, purge the dispensing system. Condition the electrodes in H_2SO_4 for four hours. Zero the meter by setting the range to 1000 millivolts (mV) according to the manufacturer's instructions.

a. Grind the potassium nitrate standard (4.4.5.2.1.b) to the consistency of table salt (easily passes through a U.S. Alternative No. 40 sieve). Pre-weigh 0.50 ± 0.05 g potassium nitrate in a weighing bottle. Dry in an oven at 150˚C for four hours. If the potassium nitrate has been out of the oven for more than four hours, re-dry at 105˚C for two hours. Store in a desiccator.

b. Set controller functions (see Table V) for a dead-stop potentiometric titration. In a hot-dry beaker, add a stir bar and 150 mL H_2SO_4. Place beaker in bath on stirrer. Ensure coolant is circulating through the external bath. Weigh potassium nitrate to the nearest 0.1 mg and slowly pour into vortex of the acid. Reweigh the empty bottle and calculate weight of potassium nitrate added. Lower the electrodes and delivery tip into acid. Turn meter to 1000 mV. When sample has dissolved, begin titration. Adjust titrant addition rate to obtain an endpoint within eight to ten minutes. After endpoint is reached and held for one minute, turn meter to standby. Rinse electrodes in H_2SO_4. Calculate the nitrogen equivalence factor (F):

$$\text{Factor} = \frac{13.85 \times W}{V}$$

Where: F = factor, % N/mL
W = weight of potassium nitrate, g
V = volume of titrant, mL

TABLE V. Recommended instrument settings.

Meter	Temperature: 25˚C Function: standby or 1000 mV
Endpoint	350 mV
Prop band	85
Min del	0

4.4.5.2.4 Procedure. Grind a portion of potassium nitrate sample to the consistency of table salt (easily passes through a U.S. Alternative No. 40 sieve). Weigh 0.50 ± 0.05 g of sample in a weighing bottle. Dry at 150˚C for two hours. Repeat 4.4.5.2.3.b with the sample potassium nitrate. Calculate the nitrogen content of the sample potassium nitrate as follows:

$$\text{Percent nitrogen} = \frac{V \times F}{W}$$

Where: F = equivalence factor, % N/mL (see 4.4.5.2.3)

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W = weight of potassium nitrate sample, g
V = volume of titrant, mL

Perform test in triplicate. Results shall be within 0.03% or the sample shall be classed defective. Calculate and report the arithmetic mean of the three values for conformance.

4.4.6 Chlorides. Chlorides content shall be determined by one of the following methods.

4.4.6.1 Chlorides by ion chromatography (preferred). The following procedure is used to determine the potassium chloride content. The potassium nitrate sample is dissolved in deionized water and the solution analyzed for chloride by suppressed anion chromatography with conductivity detection. The chloride content is determined by comparison to a known standard.

4.4.6.1.1 Reagents.

a. Deionized or distilled water.
b. Sodium chloride, ACS grade.
c. Chromatography eluent, an aqueous solution in deionized water, typically containing 2 to 6 millimolar sodium carbonate and 1 to 2 millimolar sodium bicarbonate. The exact ratio depends on chromatography column (see 4.4.6.1.2.d); a low concentration of carbonate yields slower elution but better resolution while a high concentration of carbonate yields faster elution but poorer resolution.
d. Suppressor regenerating solution as recommended by chromatograph manufacturer, typically a 50-100 millimolar solution of sulfuric acid in deionized water.

4.4.6.1.2 Supplies and equipment.

a. Analytical balance, 200 gram capacity, 0.0001 gram resolution or better.
b. Oven, set to 150±2˚C
c. Ion chromatograph with anion suppression and conductivity detector with minimum sensitivity of 100 microsiemens/centimeter full scale.
d. Ion chromatography column, which together with the proper eluent (4.4.6.1.1.c) can baseline-resolve chloride and completely elute sulfate in 30 minutes or less, e.g. Metrohm Metrosep ASupp 5 150 x 4 millimeters 5 micron, or equivalent.
e. Volumetric flasks, quantity: 1 x 1 L, 2 x 250 mL, 2 x 100 mL
f. Volumetric pipet, 1 mL, Class A.
g. Prior to starting, all flasks shall be rinsed at least three (3) times with deionized water and allowed to air dry.

4.4.6.1.3 Calibration.

a. Weigh and accurately record 0.1650 ± 0.0050 grams of sodium chloride into a 1 L volumetric flask and make up to volume with deionized water and mix well.
b. Transfer a 1 mL aliquot of the stock solution via volumetric pipette to a 100 mL volumetric flask. Similarly, transfer 2x 1 mL aliquots of the stock solution to a second 100 mL flask. Make both flasks up to volume with deionized water and mix well. Calculate the exact

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concentrations, in milligrams per liter, of potassium chloride equivalent for both standards, C1 and C2, respectively, as:

$$C1 = 12.8 \times W_{NaCl}$$
$$C2 = 25.5 \times W_{NaCl}$$

Where: W_{NaCl} is the weight of sodium chloride, grams

4.4.6.1.4 Sample preparation. The chloride analysis shall be performed on two replicate aliquots, 0.815 gram each, of potassium nitrate using the following steps; alternately solutions previously prepared for nitrogen content determination (4.4.5.1.4) may be used instead. If both nitrogen and chloride are to be determined by ion chromatography (per 4.4.5.1 and 4.4.6.1, respectively) using the same instrument, nitrogen analysis shall be run first to avoid nitrate carryover from the concentrated samples used in the chloride determinations.

a. Crush a 5 ± 1 g of the sample to pass through a U.S. Alternative No. 100 sieve conforming to ASTM E11, and dry for 2 hours in a 150±2˚C oven. Cool in a desiccator.

b. Weigh 0.8150 ± 0.0050 grams, and record exact weight, of the dried potassium nitrate into a 250 mL volumetric flask. Fill to about half volume with deionized water and swirl to dissolve. Make up to volume with deionized water and mix well.

c. Repeat 4.4.6.1.4.b to prepare the second replicate.

4.4.6.1.5 Procedure.

a. Set up the ion chromatograph in accordance with manufacturer instructions. The instrument shall be equilibrated until a steady, linear baseline is obtained. It is recommended to inject 1-3 samples of deionized water while the instrument is equilibrating to make sure the chromatograms do not contain any analytes other than water and/or system peaks.

b. Fill up the appropriate instrument vials, one each, with the chloride standards (4.4.6.1.3.b), and the two sample solutions (4.4.6.1.4.c).

c. All solutions shall be analyzed using a method appropriate for the selected chromatography column (4.4.6.1.2.d) and system. Allow each run to be just long enough (thirty minutes minimum) to completely elute any sulfate which may interfere with integration of the subsequent sample.

d. Identify the chloride peak, which elutes during the first half of each run. For the standards, chloride shall be the largest peak in the chromatogram; note its retention time. For the samples, examine the same retention time ± 5%; chloride shall be identified as the largest peak in the same retention time window.

e. Integrate each chromatogram using valley-to-valley type integration making sure the start and end points of the chloride peak both coincide with the local baseline minima. Obtain the chloride peak areas for each of the runs.

f. For the standards (4.4.6.1.3.b), obtain a linear calibration of peak area versus potassium chloride concentration (or vice versa) using a spreadsheet or software included with the instrument.

g. From the calibration line, read off the potassium chloride concentration (in milligrams per liter) corresponding to the chloride peak area for each of the sample replicates (4.4.6.1.4.c).

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h. Calculate percent potassium chloride, % KCl, for each sample as follows:

$$\% \text{ KCl} = 0.025 \times \frac{S_{KCl}}{W_S}$$

Where: S_{KCl} = concentration of potassium chloride, mg/L (4.4.6.1.5.g)
W_S = sample weight, g (4.4.6.1.4.b)

i. Calculate and report the arithmetic mean of the two % KCl values found in 4.4.6.1.5.h for conformance.

4.4.6.2 Chlorides by turbidity (alternate). Determine chloride content in accordance with ACS Reagent Chemicals, Chloride Method for potassium nitrate.

4.4.6.3 Chlorides by precipitation (alternate). Dissolve an accurately weighed portion of approximately 5 g of the sample in 100 mL of distilled water and make the solution slightly acidic with nitric acid. Add a few drops of approximately 0.25N silver nitrate solution. If more than a faint opalescence results, add more silver nitrate solution until precipitation is complete. Boil for 2 minutes with constant stirring, and allow the precipitate to settle overnight in a dark place. Decant the solution through a tared filtering crucible of medium porosity. Wash the precipitate four times by decantation, using 20 mL portions of water containing 1 mL of concentrated nitric acid and 2 drops of silver nitrate solution per 100 mL. Transfer the precipitate to the crucible with 20 mL of the same wash water, wash it with 20 mL of 1 percent nitric acid solution and with two 20 mL portions of distilled water. Dry the crucible and precipitate for 4 hours at 135°C, cool in a desiccator, and weigh. Calculate the percent chlorides, as potassium chloride KCl, as follows:

$$\text{Percent KCl} = \frac{52 \times A}{W}$$

Where: A = weight of precipitate, in g
W = weight of sample, in g

4.4.7 Chlorates and perchlorates. Chlorate and perchlorate content shall be determined in accordance with the following methods. Add the perchlorate content determined in 4.4.7.1 to the chlorate content determined in 4.4.7.2 and report for conformance.

4.4.7.1 Determination of perchlorates by perchlorate-selective electrode .

4.4.7.1.1 Reagents.

a. Water, deionized (DI) or distilled.
b. Perchlorate (ClO_4^-) 1000 mg/L standard solution. Available from electrode manufacturer or prepared by dissolving 1.393 ± 0.005 g reagent grade potassium perchlorate in enough water to make 1.000 L of solution. 100 mg/L and 10 mg/L perchlorate standard solutions may be prepared by performing successive ten-fold dilutions.

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c. Reference fill solution for perchlorate electrode, 0.1M ammonium sulfate, $(NH_4)_2SO_4$ or as recommended by electrode manufacturer.

d. Ionic Strength Adjuster (ISA), 2M ammonium sulfate, $(NH_4)_2SO_4$ or as recommended by electrode manufacturer.

4.4.7.1.2 Supplies and equipment.

a. Thermo Scientific Orion Dual Star pH/ISE meter, or equivalent, equipped with an immersible stirrer and temperature sensor.

b. Perchlorate Ion Selective Electrode (ISE), double junction, glass, e.g. Oakton by Cole-Parmer, (Fisher Scientific catalog number 50-196-1299) or equivalent. Ensure that the electrode is within its service date; replace if it is past its expiration date.

4.4.7.1.3 Calibration of electrode. Refer to the ISE meter manual for the calibration procedure. The electrode shall be recalibrated within 12 hours of expected use or if ambient temperature changes by 1˚C or more during use.

a. Prepare the electrode as follows: Connect the electrode to the ISE meter. Remove the cover from electrode tip and open the side fill port. Fill the electrode to the bottom of the fill port with reference fill solution (0.1M ammonium sulfate). Soak the electrode tip in water for approximately 30 minutes.

b. Dilute 25.0 mL of 1000 mg/L perchlorate standard to 250 mL with water. Mix well to yield the 100 mg/L perchlorate standard.

c. Dilute 25.0 mL of the 100 mg/L perchlorate standard to 250 mL with water. Mix well to yield the 10 mg/L perchlorate standard.

d. Dilute 25.0 mL of the 10 mg/L perchlorate standard to 250 mL with water. Mix well to yield the 1 mg/L perchlorate standard.

e. Measure 100 ± 0.5 mL of each of the above calibration solutions into three 150 mL beakers. Pipet 2.0 mL of ISA into each beaker.

f. Immerse the electrode, stirrer, and temperature sensor into the 1 mg/L perchlorate solution and start the stirrer at a speed that does not generate a vortex. When the meter indicates a stable reading, record the voltage (mV) and temperature (˚C) of the solution. Rinse the electrode with water, and then blot it dry. Do not wipe or rub the sensing membrane on the tip of the electrode.

g. Similarly, measure the 10 mg/L and 100 mg/L perchlorate standards. Record the voltage and temperature of the solutions.

h. Save the calibration.

i. Use the instrument's plot function to determine the slope. This slope is dependent on temperature and at room temperature (~25˚C) should be in the range of -52 to -59 mV/(mg/L). If the slope is not within this range, repeat the calibration. If the calibration remains out of the specified slope range, repeat it with fresh calibration solutions.

Note: the calibration plot of voltage is linear with respect to the logarithm of concentration. All calculations are made by the ISE meter, but may be verified manually.

k. Following the calibration, rinse the electrode with water, then blot it dry. Do not wipe or rub the sensing membrane on the tip of the electrode.

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4.4.7.1.4 Sample preparation and analysis. Before proceeding with sample determinations, ensure that the electrode is filled with reference fill solution.

a. Sample solution: Weigh 2500 ± 50 mg of potassium nitrate sample into a 250 mL volumetric flask. Add 100 mL of water then sonicate and swirl until dissolved. Dilute to volume with DI water and mix.

b. Transfer 100 mL of sample solution into a 150 mL beaker. Add 2.0 mL of ISA to the beaker. Immerse the electrode, stirrer and temperature sensor into the solution and measure the concentration of perchlorate as per the meter operating manual, recording the reading when the meter has stabilized. Record the measurement and temperature.

Note: Temperatures of calibration and sample solutions should be within 1°C. If the sample solution is more than 1°C different from the calibration solutions, allow the sample solution to equilibrate until it is within 1°C of the calibration solutions. Alternately, recalibrate at the sample temperature.

c. Calculate the percentage of potassium perchlorate in the potassium nitrate by the following formula:

$$\% \, KClO_4 = \frac{1.393 \times C \times V}{W} \times 100$$

Where: 1.393 = ratio of molecular weights of $KClO_4$ and ClO_4^-,
C = measured perchlorate concentration, mg/L
V = sample volume, L
W = KNO_3 sample weight, mg

4.4.7.2 Determination of chlorate content by color comparison method.

4.4.7.2.1 Reagents and apparatus.

a. Water, deionized or distilled
b. Potassium chlorate, KClO3, ACS reagent grade
c. Potassium nitrate, KNO3, ACS reagent grade
d. Hydrochloric Acid, reagent grade, 34-37 %wt. HCl
e. Nitric acid, reagent grade, 67-70 %wt. HNO3
f. Aniline hydrochloride, 99%
g. Aniline hydrochloride reagent: Dissolve 25 grams of aniline hydrochloride in 350 mL of concentrated hydrochloric acid in a 500 mL volumetric flask. Dilute to volume with water. Activate the reagent by addition of one crystal (or about 1 mg) of potassium chlorate. Store the solution overnight, protected from light.
h. Potassium chlorate stock solution: Place 500 ± 3 mg of potassium chlorate into a 200 mL volumetric flask. Dissolve in and dilute to volume with water. (Solution concentration is approximately 2.5 mg $KClO_3$/mL).
i. Nitric acid (1:1 by volume): Mix equal volumes of water and 67-70 weight percent reagent grade nitric acid.
j. Flasks as needed.
k. Color comparison tube rack for 50 mL tubes

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l. Pipets, 2x 10 mL and 2x 20 mL (individual “To deliver” or serological/graduated)

Note: 100 mL and 125 mL flasks and color comparison tubes shall be rinsed with 1:1 nitric acid/water to remove interferences, then rinsed with water, and dried before use.

4.4.7.2.2 Preparation of potassium chlorate standard solutions.

a. Place 5.00 ± 0.05 g potassium nitrate (ACS reagent) into each of six 100 mL volumetric flasks. Add 20 mL water to each, then sonicate and mix until dissolved.

b. Add the indicated amount of potassium chlorate stock solution to each individual 100 mL flask per Table VI:

TABLE VI. Standard solutions.

Flask	Potassium chlorate stock, mL	% $KClO_3$ on KNO_3
1	0.0	0.0
2	2.0	0.1
3	4.0	0.2
4	6.0	0.3
5	8.0	0.4
6	10.0	0.5

4.4.7.2.3 Sample preparation and analysis.

a. Weigh 5.00 ± 0.05 g of the potassium nitrate sample into a 100 mL volumetric flask. Add 20 mL of water. Sonicate and mix until dissolved.

b. Add 20.0 mL aniline HCl reagent to the standard flasks 1 - 6 and the sample flask. Swirl to mix. Dilute to volume with water and mix. Solutions in flasks 2 to 6 should turn blue. Flocculent precipitates may develop.

c. Filter each preparation through qualitative filter paper into conical 125 mL flasks. Solutions in flasks 2 to 6 should be violet. The sample solution may be violet as well. Fill each solution into 50 mL color comparison tubes to the fill line. Cap the tubes and place them into a color comparison rack with an angled white reflective surface below the tubes.

d. Use a suitable light source to evenly illuminate the tubes from the white reflective surface below the tubes. Looking down into the solutions, view the tubes through the caps and compare the color intensity of the sample preparation to the standard preparations, matching the sample to the closest standards. The comparisons shall be made within one hour of the addition of the aniline HCl reagent.

e. Record which standard tube is most closely matched by the sample. If the sample tube is between two standards, record the higher of the two standards as the amount of potassium chlorate in the sample. For example, if the sample tube is between standard tubes 1 (0%) and 2 (0.1%), record 0.1% as the sample measurement.

f. Convert the amount of potassium chlorate found to its equivalent as potassium perchlorate by multiplying by the factor 1.13.

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Note: Clean the glassware of aniline residue by rinsing them with 1:1 nitric acid, then rinse with water until free of blue color. It may be necessary to repeat the acid and water rinsing procedure.

4.4.8 Trace impurities. Trace impurities content shall be determined in accordance with one of the following methods.

4.4.8.1 Iron, aluminum, calcium, magnesium, and sodium by Inductively Coupled Plasma – Optical Emission Spectroscopy (ICP-OES) method (preferred). Class A glassware shall be used throughout this method.

4.4.8.1.1 Reagents. It is advisable to use commercially available stock solutions with certified concentrations traceable to primary standards (National Institute of Standards and Technology or international measurement standards). Stock solutions shall be replaced after one year or in accordance with manufacturer recommendations. Alternately, stock solutions may be prepared in-house from high purity metals or their salts; these materials shall be Reagent grade or better. The maximum recommended shelf life is one year from date of initial preparation when kept tightly sealed at 5-25°C. Common commercially available stock concentrations are as follows:

a. Certified Na stock solution, 100 ± 2 mg/L in 0-3% (v/v) HNO_3
b. Certified Mg stock solution, 100 ± 2 mg/L in 0-3% (v/v) HNO_3
c. Certified Al stock solution, 100 ± 2 mg/L in 0-3% (v/v) HNO_3
d. Certified Ca stock solution, 100 ± 2 mg/L in 0-3% (v/v) HNO_3
e. Certified Fe stock solution, 100 ± 2 mg/L in 0-3% (v/v) HNO_3

Additional necessary reagents are:

f. De-ionized water, conductivity <0.1 µS/cm; LC-MS (liquid chromatography – mass spectrometry) grade water may be used instead.
g. Nitric acid (HNO_3), concentrated, 67-70%, trace metal grade.
h. Nitric acid (HNO_3), 3%, made by diluting 22 ± 2 mL of concentrated nitric acid to 500 mL with deionized water; this is the minimum recommended volume of 3% nitric acid required for analysis.
i. Compressed argon, ultra-high purity (UHP-grade), as per ICP-OES manufacturer recommendations.

4.4.8.1.2 Preparation of reference (standard) solutions. Prepare 1 mg/L, 5 mg/L, and 10 mg/L standards. If using the stock concentrations listed in 4.4.8.1.1 then use the following steps.

a. Pipette 1 mL of each certified stock solution into a 100 mL volumetric flask. Fill to the mark with 3% nitric acid and stopper the flask. Invert several times to mix. This is the 1 mg/L standard.
b. Pipette 5 mL of each certified stock solution into a 100 mL volumetric flask. Fill to the mark with 3% nitric acid and stopper the flask. Invert several times to mix. This is the 5 mg/L standard.
c. Pipette 10 mL of each certified stock solution into a 100 mL volumetric flask. Fill to the mark with 3% nitric acid and stopper the flask. Invert several times to mix. This is the 10 mg/L standard.

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d. Prepare a fourth flask of 3% nitric acid and retain as the blank solution.

4.4.8.1.3 Sample preparation.

a. Weigh 1000 mg ± 50 mg of sample to the nearest mg and transfer into a 100 mL volumetric flask.
b. Add approximately 90 mL of the 3% nitric acid to the flask and place the stopper on the flask.
c. Allow the sample to completely dissolve. Agitation by means of rotary shaker or ultrasonic bath may be used if necessary.
d. Fill to the mark with 3% nitric acid and invert several times to mix.

4.4.8.1.4 Instrument preparation. Prepare the instrument according to manufacturer instructions. It is recommended to perform all measurements in radial mode. Instrument response for each analyte shall be such that the response of the 1 mg/L standard is greater than ten times that of the blank, and the response of the 10 mg/L solution does not saturate the detector. Ensure a steady baseline with no drift; measure the response of the 5 mg/L standard at 5 minute intervals until successive measurements differ by no more than 5%. Allow for adequate flush time of the system between standard and sample solutions to ensure there is no carryover. Calibrate the system for all five analytes using the standard solutions and blank. Analyze the samples in the same run, recording the results in mg/L. The wavelengths used for each element are as follows:

TABLE VII. Wavelengths for ICP-OES element analysis.

Element	Wavelength (nm)
Fe	259.940
Al	167.079
Ca	393.366
Mg	279.553
Na	589.592

4.4.8.1.5 Analysis.

a. For each element generate a four-point calibration line utilizing the prepared standards and the blank. Force the line through zero concentration. The line shall have a best fit value (R^2) of not less than 0.95.
b. Aspirate each sample for the time required by the instrument to achieve a stable response. Measure the response and calculate the concentration of each element from its corresponding calibration line, recording the results in mg/L.
c. If the calculated concentration is higher than 10 mg/L, perform a 10-fold dilution of the sample (e.g. 5 mL of original sample solution diluted to 50 mL with 3% nitric acid) and repeat previous step using the 10-fold-diluted sample. Multiply the result by 10 to determine the original concentration in mg/L.

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d. Calculate the percent compound for direct comparison to the chemical requirements in Table I as follows. Multiply the measured concentration in mg/L for each element by the corresponding conversion factor in Table VIII. Furthermore, calculate the total weight percent of aluminum oxide plus iron oxide, and the total weight percent of calcium oxide plus magnesium oxide.

TABLE VIII. Conversion factors.

Element	Chemical requirement per Table I	Conversion factor
Fe	Iron (as Fe_2O_3)	0.0143
Al	Aluminum (as Al_2O_3)	0.0189
Ca	Calcium (as CaO)	0.0140
Mg	Magnesium (as MgO)	0.0166
Na	Sodium (as Na_2O)	0.0135

4.4.8.2 Iron and aluminum (applicable to Class 1 and 2 only) (alternate). Dissolve a weighed portion of approximately 5 g of the sample in distilled water and filter the solution. Acidify the filtrate with nitric acid, add a slight excess of ammonium hydroxide, and boil. Catch the precipitate on the ashless filter paper and wash with hot distilled water. Retain the filtrate and washings for further analysis. Ignite the filter paper and precipitate, in a tared porcelain crucible, to constant weight. Cool the crucible and contents in a desiccator, and weigh. Calculate the increase in weight as percent iron and aluminum oxides.

$$\text{Percent Fe and Al oxides} = 100 \times \frac{C - D}{E}$$

Where: C = weight of precipitate and crucible, in g
D = weight of tared crucible, in g
E = weight of original sample (approx. 5 g), in g

4.4.8.3 Calcium and magnesium (alternate). Dilute the combined filtrate and washings from the determination of iron and aluminum (see 4.4.8.2) to approximately 200 mL and boil. Evaporate the filtrate to a volume of about 20 mL, filter, and add 15 mL of 9N ammonium carbonate solution, and 15 mL of 95 percent ethyl alcohol. Allow the mixture to stand for at least 30 minutes, with frequent stirring. Filter through paper and wash the precipitate with 9N ammonium carbonate solution. Dissolve the precipitate in dilute hydrochloric acid, dilute to approximately 50 mL, boil, and add 1 mL of saturated ammonium oxalate solution. Allow the solution to stand for 12 hours, filter through a tared filtering crucible, and wash the precipitate with hot water containing ammonium oxalate. Ignite the precipitate, gently at first and then with

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a blast lamp, cool the crucible and contents in a desiccator, and weigh. Calculate the increase in weight as percent calcium oxide, as follows:

$$\text{Percent calcium oxide} = \frac{100 \times A}{W}$$

Where: A = increase in weight of crucible, g
W = weight of sample, g

Concentrate the filtrate from the calcium determination to approximately 50 mL, add 10 mL of a 10 percent solution of ammonium phosphate, and 20 mL of a 10 percent solution of ammonia. Allow the solution to stand for 4 or 5 hours, filter through a tared filtering crucible, and wash with a 2.5 percent solution of ammonia. Dry the precipitate and ignite to constant weight with a blast lamp. Cool the crucible and contents in a desiccator, and weigh the magnesium pyrophosphate precipitate. Calculate the increase in weight as a percent magnesium oxide, as follows:

$$\text{Percent magnesium oxide} = \frac{36 \times A}{W}$$

Where: A = weight of magnesium pyrophosphate precipitate, g
W = weight of sample, g

4.4.9 Ammonium compounds (applicable to Class 1 and 2 only). Ammonium compound content shall be determined in accordance with the following method. Transfer an accurately weighed portion of approximately 1.0 g of the sample to a 150 mL beaker, dissolve in 25 mL of distilled water, and add sufficient excess of concentrated sodium hydroxide solution to render the potassium nitrate solution alkaline. Heat the mixture and test the vapors with a moist strip of red litmus paper of such sensitivity that it changes color in 30 seconds when immersed in 0.0005N alkali and stirred continuously. Care should be taken that no spray from the solution touches the test paper. Note if any ammonium compounds are present in the sample, as indicated by the paper becoming blue in color.

4.4.10 Granulation.

4.4.10.1 Class 1, lump. The sample shall be examined visually for conformance.

4.4.10.2 Classes 2, 3, and 4. Granulation shall be determined in accordance with the following method. Nest the specified sieves complying with ASTM E11 on a bottom pan. Place a weighed portion of 100 grams of the sample on the upper sieve, cover, and shake for ten minutes by hand, or for five minutes by means of a mechanical shaker geared to produce 268 to 315 gyrations and 150 ± 10 taps of the striker per minute. Weigh the amounts retained by the sieves, and calculate to percent as required.

4.4.11 Workmanship. Workmanship shall be verified in accordance with the following method. Take approximately 300 grams of the material and spread the composition out on a

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clean white surface. Verify by visual and nasal examination that the sample is uniform in quality, white in color, free of odor, and free from lumps (Classes 2, 3, and 4), dust, debris, visible impurities, or foreign matter.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in contract or order (see 6.2). When packaging of materiel is to be performed by DOD or in-house personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. Commercial-grade potassium nitrate is intended for use in ammunition.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Class of material required (see 1.2).
c. Packaging requirements (see 5.1).
d. Requirements for First article samples (see 4.2).
e. Provision for submission of first article samples.
f. Requirements for submission of acceptance inspection equipment.
g. Requirements for ammunition lot numbering.

6.3 Submission of contractor acceptance inspection equipment (AIE) designs for approval. Submit copies of designs as required to the Technical Agency (see 6.5).

6.4 Equivalent test method approval. Prior approval of the contracting officer is required for use of equivalent test methods. A description of the proposed method should be submitted to the Technical Agency (see 6.5). This description should include but not be limited to the procedures used, the accuracy and precision, and drawings of any special equipment required.

6.5 Technical agency. Picatinny Arsenal is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: DEVCOM Armaments Center, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000.

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6.6 Changes incorporated. This specification was updated from MIL-DTL-156C to MIL-DTL-156D with Amendment 1 as part of the Munitions Chemical Specification Modernization (MCSM) program. All test procedures were verified and updated as needed to produce valid results. Reference documents have been verified, updated, and hyperlinked to sources. Sampling procedures have been updated. Details of these changes are documented in the ECPs and ERR listed below.

Specification ID	ECP / ERR
MIL-P-156B	-
MIL-DTL-156C	Y18Q2003
MIL-DTL-156D	Y22Q2003
MIL-DTL-156D w/ Amendment 1	Y24Q3010

6.7 Subject term (key word) listing.

Black powder
Propellant
Pyrotechnic
Munitions Chemical Specification Modernization (MCSM)

6.8 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodians:
Army – AR
Navy – OS
Air Force – 170

Preparing Activity:
Army – AR
(Project: 6810-2025-011)

Review Activities:
Army – AV, TE
Navy – AS, SH
Air Force – 11, 70

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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